

AWS SUMMIT



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Supercharge your productivity across the SDLC with generative AI

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Agenda

- 01 Understanding the development lifecycle
- 02 A deep dive into Amazon Q Developer
- 03 Demos



Improving
productivity with
generative AI

30%

Productivity gains

75%

Total use cases

Use cases

Software engineering
Customer operations

Research and development
Marketing and sales

Where are developers spending time in the SDLC?

Plan

- Find accurate and specific technical guidance from relevant documentation, code examples, and other resources
- Design changes based on business and technical requirements

Create

- Generate code
- Manage infrastructure

Test and secure

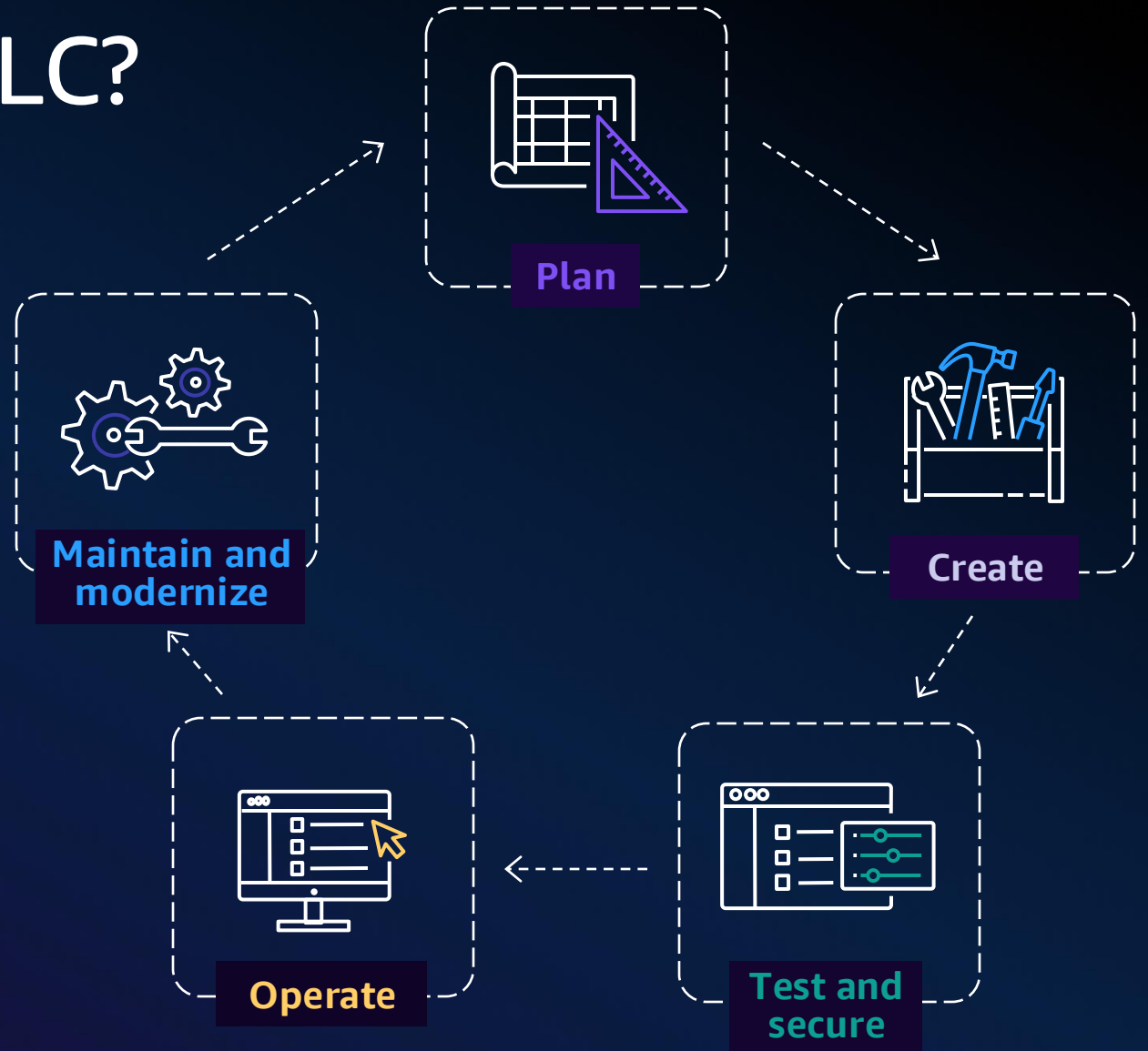
- Apply proper test cases for code changes
- Scan for security vulnerabilities in code

Operate

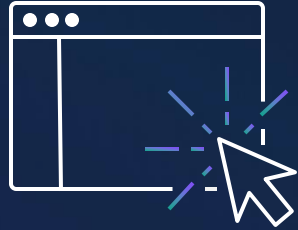
- Identify and mitigate code issues
- Monitor performance and efficiency

Maintain and modernize

- Modernize and update old code languages and dependencies



Developer challenges beyond code authoring



**Onboarding to
existing projects**



**Setting up app
configuration**



**Broadening
test coverage**



**Deploying and
operating the app**



Amazon Q transforms
the **development** experience

Amazon Q Developer



Helps developers and IT professionals build faster across the entire software development lifecycle (SDLC)

Most accurate coding recommendations

Agents can autonomously help you implement features, refactor code, perform software upgrades and more

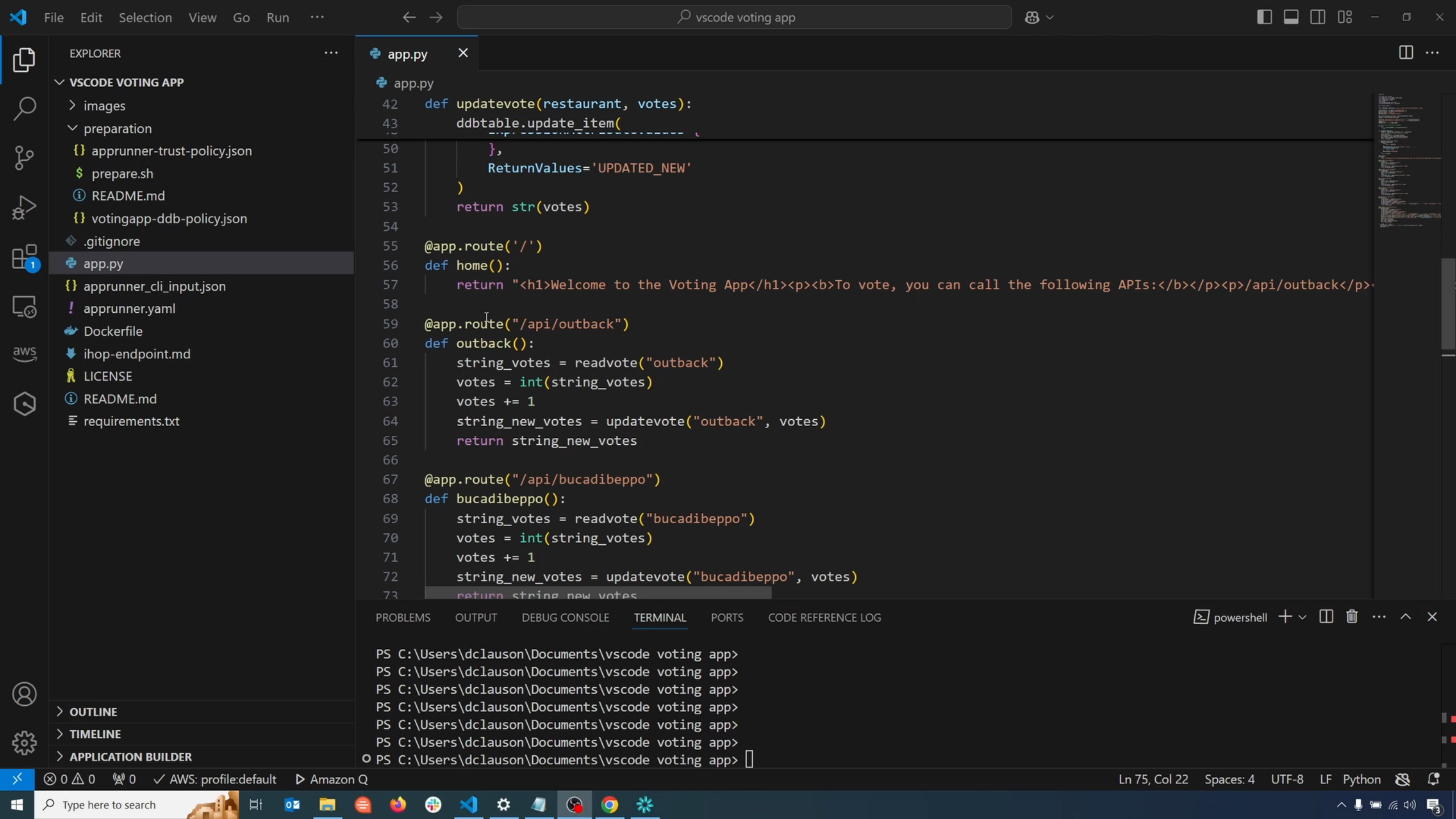
Amazon Q is an expert on AWS and optimizing AWS environment

Best-in-class security vulnerability scanning and remediation

Amazon Q is built with security and privacy in mind from the start, making it easier for organizations to use generative AI safely.

Plan and create







**SOFTWARE
DEVELOPMENT**



TRANSFORMATION



QUALITY ASSURANCE



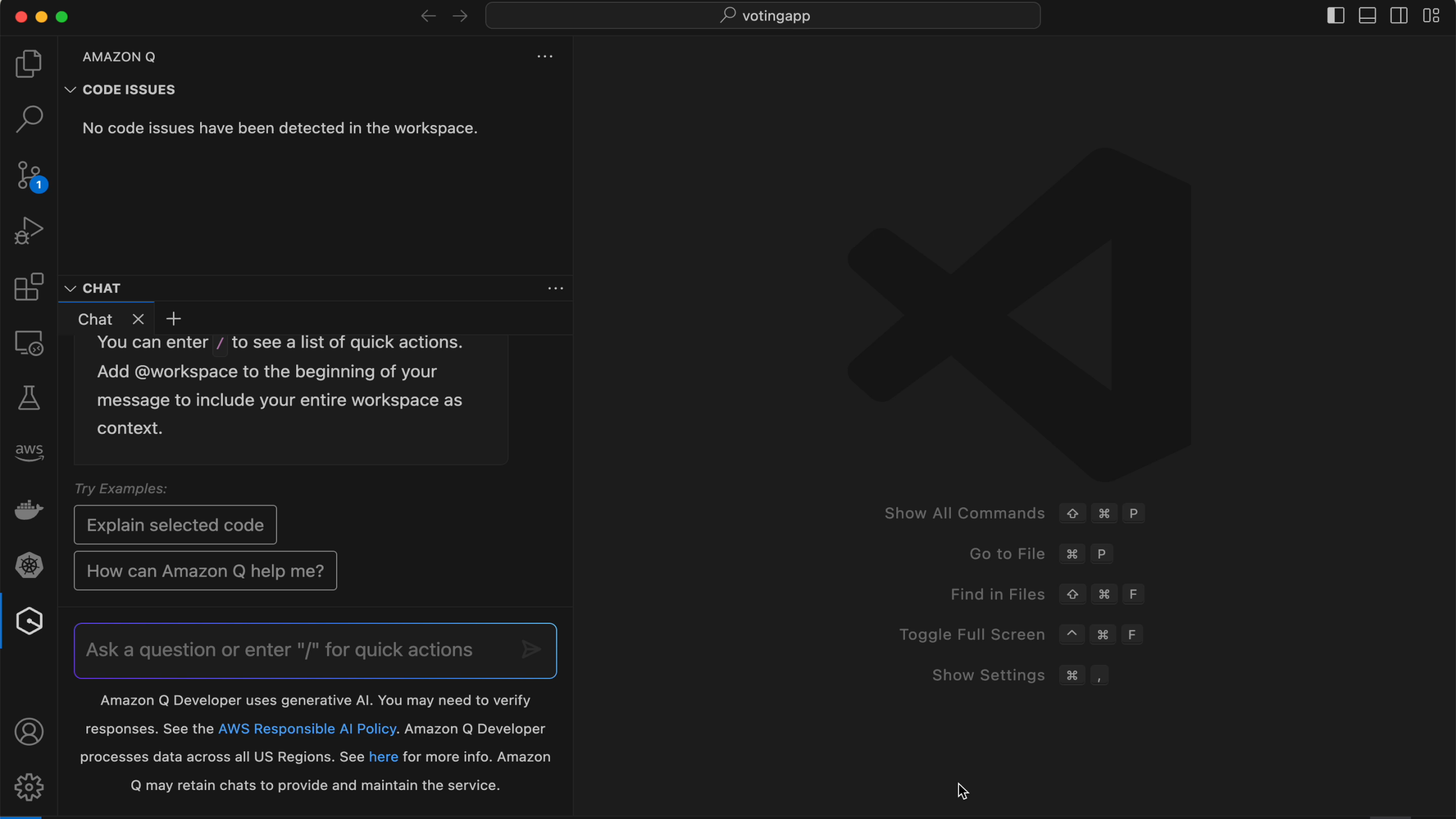
OPERATIONS



OPTIMIZE

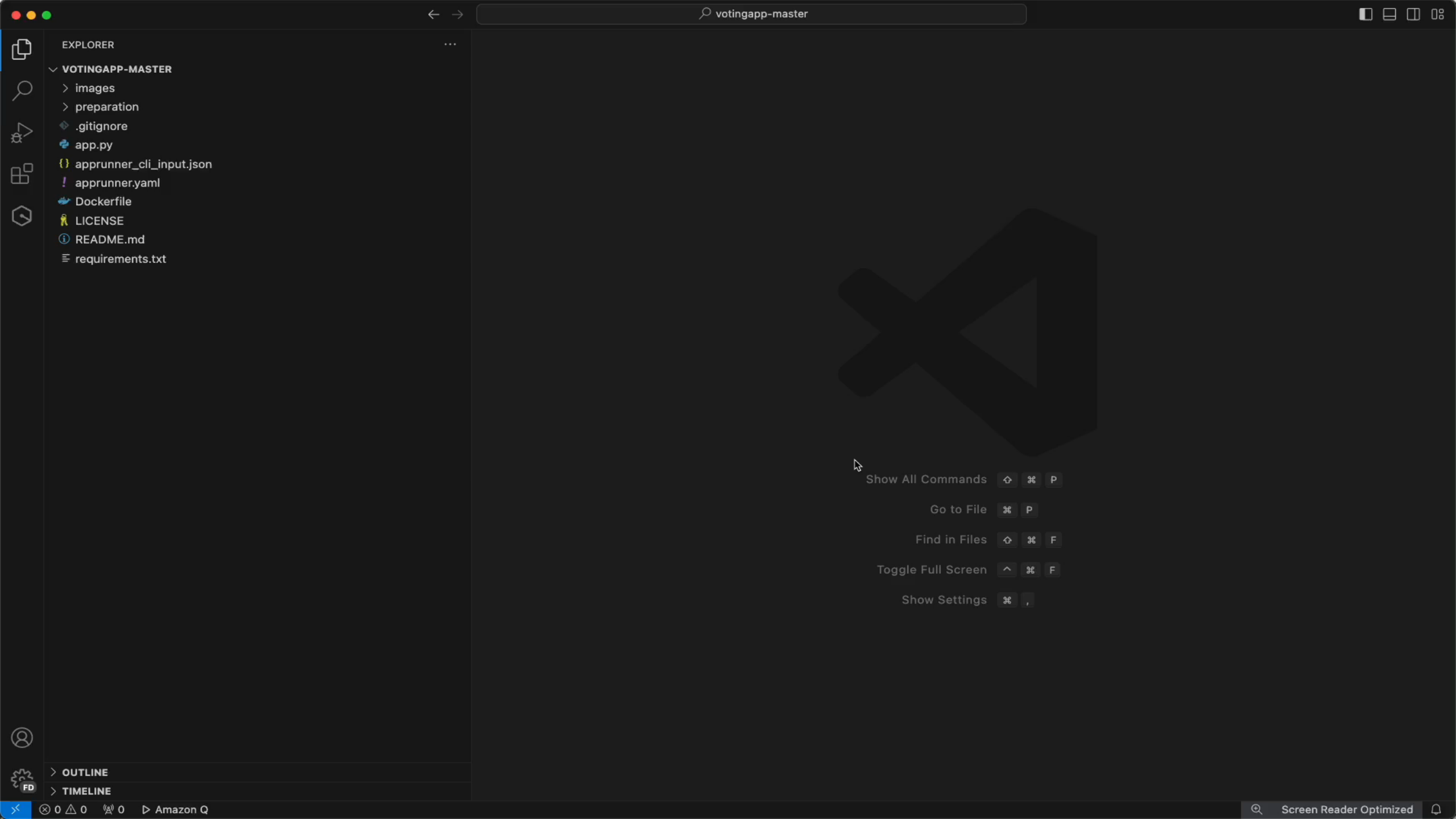
Agentic Everything





Test and operate





EXPLORER

- ▼ VOTINGAPP-MASTER
 - > images
 - > preparation
 - 📄 .gitignore
 - 📄 app.py
 - 📄 apprunner_cli_input.json
 - 📄 ! apprunner.yaml
 - 📄 Dockerfile
 - 📄 LICENSE
 - 📄 README.md
 - 📄 requirements.txt

- > OUTLINE
- > TIMELINE



- Show All Commands ⌘⇧ P
- Go to File ⌘⇧ P
- Find in Files ⌘⇧ F
- Toggle Full Screen ⌘⇧ F
- Show Settings ⌘ ,

AMAZON Q

CODE ISSUES

No code issues have been detected in the workspace.

CHAT

Chat × +

your entire workspace as context.

Try Examples:

Explain selected code

How can Amazon Q help me?

/



Amazon Q Developer uses generative AI. You may need to verify responses. See the [AWS Responsible AI Policy](#). Amazon Q Developer processes data across all US Regions. See [here](#) for more info. Amazon Q may retain chats to provide and maintain the service.

app.py M X

app.py > updatevote

```
35 def readvote(restaurant):
39     json_response = json.loads(normalized_response)
40     votes = json_response["Item"]["restaurantcount"]
41     return str(votes)
42
43 def updatevote(restaurant, votes):
44     ddbtable.update_item(
45         Key={
46             'name': restaurant
47         },
48         UpdateExpression='SET restaurantcount = :value',
49         ExpressionAttributeValues={
50             ':value': votes
51         },
52         ReturnValues='UPDATED_NEW'
53     )
54     return str(votes)
55
56 @app.route('/')
57 def home():
58     return "<h1>Welcome to the Voting App</h1><p><b>To vote, you can call the following APIs:</b></p><p>/a
59
60 @app.route("/api/outback")
61 def outback():
62     string_votes = readvote("outback")
63     votes = int(string_votes)
64     votes += 1
65     string_new_votes = updatevote("outback", votes)
66     return string_new_votes
67
68 @app.route("/api/bucadibeppo")
69 def bucadibeppo():
70     string_votes = readvote("bucadibeppo")
71     votes = int(string_votes)
72     votes += 1
```

Stacks (1)



Filter status

Active



View nested

< 1 >

Stacks

cloudformation-beb308e256

2024-10-16 04:46:06 UTC-0700

 CREATE_FAILED

cloudformation-beb308e256



Delete

Update

Stack actions

Create stack

 Stack rollback paused

All successfully provisioned resources are live. View the Events tab and remediate any resource errors to continue. Select Retry if the resource failed to provision due to an issue that doesn't require template modifications. Select Update if the resource failed to provision due to template errors. Select Roll back to roll back to the last known stable state. [Learn more](#)

Retry

Update

Roll back

Stack info

Events - updated

Resources

Outputs

Parameters

Template

Change sets

Git sync

Table view

Timeline view - new

 The resource MyLambdaFunctioncloudformationbeb308e256E357C6A3 is in a CREATE_FAILED state Diagnose with Amazon Q

Events (12)

View root cause



Timestamp	Logical ID	Status	Detailed status	Status reason
2024-10-16 04:46:36 UTC-0700	cloudformation-beb308e256	 CREATE_FAILED	-	The following resource(s) failed to create: [MyLambdaFunctioncloudformationbeb308e256E357C6A3].
2024-10-16 04:46:35 UTC-0700	MyLambdaFunctioncloudformationbeb308e256E3	 CREATE_FAILED	-	Resource handler returned message: "Uploaded file must be a non-empty zip (Service: Lambda, Status Code: 400, Request ID: 55c793e4-00c7-4f23-...)

Likely root cause

Untitled job

Actions Save Run

ScriptJob detailsRunsData qualitySchedulesVersion Control

ScriptInfo

```
1 import sys
2 from awsglue.transforms import *
3 from awsglue.utils import getResolvedOptions
4 from pyspark.context import SparkContext
5 from awsglue.context import GlueContext
6 from awsglue.job import Job
7
8 ## @params: [JOB_NAME]
9 args = getResolvedOptions(sys.argv, ['JOB_NAME'])
10
11 sc = SparkContext()
12 glueContext = GlueContext(sc)
13 spark = glueContext.spark_session
14 job = Job(glueContext)
15 job.init(args['JOB_NAME'], args)
16 job.commit()
```

Amazon Q

Why can't the EC2 instance web01 connect to the RDS instance demo?

It looks like you need help with network connectivity issues. Amazon Q works with VPC Reachability Analyzer to provide an interactive generative AI experience for troubleshooting network connectivity issues. You can try the [preview experience here](#) (available in US East N. Virginia Region).

LikeQuoteShare

II

New conversation

What is the difference between a Spark DataFrame and a Glue DynamicFrame?

3927 characters left

Use of Amazon Q is subject to the [AWS Responsible AI Policy](#)

PythonLn 1, Col 1Errors: 0Warnings: 0

Maintain and modernize



Amazon Q Developer transformation capabilities

VMware

.NET

Mainframe

Java



8



9

Search or go to...

Project

download-server

Pinned

Issues

0

Merge requests

1

Manage

Plan

Issues

0

Issue boards

Milestones

Iterations

Wiki

Requirements

Code

Build

Secure

Deploy

Operate

Monitor

Analyze

Settings

Help

Admin

New Issue

Title (required)

Type ?

Issue

Description

Preview

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[Link](#)

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Write a description or drag your files here...

Switch to rich text editing

Id description email address your contact information in it effectively!

☐ This issue is confidential and should only be visible to team members with at least Reporter access.

Assignees

Unassigned

Assign to me

Epic

Select epic

Milestone

Select milestone

Labels

Select label

Weight

Enter a number

Due date

Select due date

Iteration

Select iteration

Create issue

Cancel

Hot off the press!





Amazon Q Developer CLI Agent

```
/tmp % q chat --accept-all
```

Hi, I'm **Amazon Q**. Ask me anything.

Things to try

- Fix the build failures in this project.
- List my s3 buckets in us-west-2.
- Write unit tests for my application.
- Help me understand my git status

/acceptall	Toggles acceptance prompting for the session.
/help	Show the help dialogue
/quit	Quit the application

```
> Scaffold a new application named call-for-content using React and  
en commit it to Git.█
```


MCP - USB Port for AI?



Get started with Amazon Q Developer



**Try Amazon Q
Developer today**



skillbuilder.aws 

Create a free account
on AWS Skill Builder to
gain in-demand skills

Thank you!

Senthilkumar Jeyachandran

AI & ML Specialist Solutions Architect
AWS India



Please complete the session
survey in the mobile app

